

< Previous







Next >

1. Inverses and solving linear systems

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Part B due Feb 14, 2024 20:30 IST

Find the inverse

4/4 points (graded)

Find the inverse of the matrix

$$\mathbf{D} = \begin{pmatrix} 7 & 2 & 1 \\ 0 & 3 & -1 \\ -3 & 4 & -2 \end{pmatrix}.$$

(Enter a matrix in python array notation. That is, enter rows between square brackets separated by commas, separate rows by commas, and surround the entire matrix with square brackets; e.g. type

`[[1, 0, 0], [0, 1, 0], [0, 0, 1]]` for the 3×3 identity matrix. Note that it will render correctly below the answer box if typed correctly.)

$\mathbf{D}^{-1} =$

`[-2,8,-5],[3,-11,7],[9,-34,21]` ✓ Answer: `[-2, 8, -5],[ 3, -11, 7], [9, -34, 21]]`

Find a solution to the following system of equations.

$$7x_1 + 2x_2 + x_3 = 21$$

$$3x_2 - x_3 = 5$$

$$-3x_1 + 4x_2 - 2x_3 = -1$$

$x_1 =$ ✓ Answer: 3

$x_2 =$ ✓ Answer: 1

$x_3 =$ ✓ Answer: -2

Solution:

Find the inverse of $\mathbf{D}$ by hand or with MATLAB. We will illustrate how to do it by hand here.

$$\left[\begin{array}{ccc|ccc} 7 & 2 & 1 & 1 & 0 & 0 \\ 0 & 3 & -1 & 0 & 1 & 0 \\ -3 & 4 & -2 & 0 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 7 & 2 & 1 & 1 & 0 & 0 \\ 0 & 3 & -1 & 0 & 1 & 0 \\ 0 & 34/7 & -11/7 & 3/7 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 7 & 2 & 1 & 1 & 0 & 0 \\ 0 & 3 & -1 & 0 & 1 & 0 \\ 0 & 34 & -11 & 3 & 0 & 7 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 7 & 2 & 1 & 1 & 0 & 0 \\ 0 & 3 & -1 & 0 & 1 & 0 \\ 0 & 0 & 1/3 & 3 & -34/3 & 7 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 7 & 2 & 1 & 1 & 0 & 0 \\ 0 & 3 & -1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 9 & -34 & 21 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 7 & 2 & 0 & -8 & 34 & -21 \\ 0 & 3 & 0 & 9 & -33 & 21 \\ 0 & 0 & 1 & 9 & -34 & 21 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 7 & 2 & 0 & -8 & 34 & -21 \\ 0 & 1 & 0 & 3 & -11 & 7 \\ 0 & 0 & 1 & 9 & -34 & 21 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 7 & 0 & 0 & -14 & 56 & -35 \\ 0 & 1 & 0 & 3 & -11 & 7 \\ 0 & 0 & 1 & 9 & -34 & 21 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -2 & 8 & -5 \\ 0 & 1 & 0 & 3 & -11 & 7 \\ 0 & 0 & 1 & 9 & -34 & 21 \end{array} \right]$$

Therefore $\mathbf{D}^{-1} = \begin{pmatrix} -2 & 8 & -5 \\ 3 & -11 & 7 \\ 9 & -34 & 21 \end{pmatrix}.$

The solution to the system of linear equations is given by

The solution to the system of linear equations is given by

$$\mathbf{D}^{-1} \begin{pmatrix} 21 \\ 5 \\ -1 \end{pmatrix} = \begin{pmatrix} -2 & 8 & -5 \\ 3 & -11 & 7 \\ 9 & -34 & 21 \end{pmatrix} \begin{pmatrix} 21 \\ 5 \\ -1 \end{pmatrix} = \begin{pmatrix} -42 + 40 + 5 \\ 63 - 55 - 7 \\ 189 - 170 - 21 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ -2 \end{pmatrix}.$$

Alternatively, you can use the augmented matrix to solve this system directly. This is more efficient if you do not need to find the inverse matrix because solving for the inverse requires computing 9 numbers, while solving the system only computes 3 numbers.

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More about inverses

2/2 points (graded)
(In the following two problems, assume that **A** is a matrix, and **x** and **b** are column vectors.)

Which of the following must be true for **Ax = b** to imply that **x = A⁻¹b**?

- ☐ **A** is singular.
- ☒ **A** is nonsingular.
- ☒ **A** is a square matrix.
- ☐ The nullspace of **A** is 1 dimensional.
- ☒ The nullspace of **A** is zero dimensional.
- ☐ The matrix **A** is **1 × 1**.
- ☐ None of the above.



Which of the following must be true for **Ax = b** to imply that **x = bA⁻¹**?

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- ☒ **A** is a square matrix.
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- ☒ The nullspace of **A** is zero dimensional.
- ☒ The matrix **A** is **1 × 1**.
- ☐ None of the above.



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Solution:

In order for $\mathbf{Ax} = \mathbf{b}$ to imply that $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$, it must be true that $\mathbf{A}$ is invertible. Therefore $\mathbf{A}$ must be square and nonsingular. A nonsingular matrix is equivalent to $\mathbf{A}$ having a zero dimensional nullspace.

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